Sourav Snigdha Mansingh

Senior Software Engineer - Java - Scalable Distributed Systems

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Professional Summary

A Senior Software Engineer having **11 years of backend development experience** with deep expertise in designing and scaling **production-grade distributed systems** using **Goland and Java**.
Proven track record of building high-impact web applications and microservices in **fast-paced, startup-style environments**, with a strong sense of ownership and responsibility for shipping features to production.
Experienced in developing systems that process **1M+ transactions per day with sub-10ms latency and 99.99% availability**, as well as **real-time SaaS(Software as a Software) applications** supporting **2M+ daily active users and 300K messages per second**.
Extensive hands-on experience with **Apache Cassandra, PostgreSQL, Redis, and Kafka, leveraging event-driven architectures, strong consistency models, and concurrency control to achieve 10× throughput improvements and 35% infrastructure cost reduction**.

Professional Experience

Confluent – Senior Software Engineer **Remote, India**
09/2024 - 06/2025

- Architected and implemented a high-scale, **real-time SaaS platform from 0 to MVP and then scaled up to a point where it served 15M DAU** across 8 global regions, sustaining 350K messages/sec with 99.95% uptime and 42ms p95 latency, using **Java 21, Spring Boot, Golang, Gin, Apache Kafka, PostgreSQL (CloudSQL), Redis, Docker, and Kubernetes (GKE).**
- Built a secure hierarchical multi-tenant architecture using **PostgreSQL partitioning, Row-Level Security (RLS), composite indexing, connection pooling (HikariCP)**, and **Spring Security with JWT/OAuth2**, scaling to 10K+ companies with strict tenant isolation.
- Built a low-latency WebSocket service in **Go** using **Gorilla WebSocket and net/http**, leveraging **goroutines, channel multiplexing, sync.Mutex/RWMutex** coordination, and errgroup-based concurrency patterns to efficiently handle high levels of parallel connections.
- Led system-wide performance optimizations including **JVM tuning (G1GC/ZGC), thread pool tuning, async processing (CompletableFuture), batching strategies, Kafka producer acks tuning, and database query optimization**, reducing p95 latency by 77% and increasing DB throughput from 5K to 50K msg/sec.
- Implemented production-grade observability and reliability using **Prometheus, Grafana, OpenTelemetry, distributed tracing (Jaeger), structured logging (ELK stack)**, and alerting via Alertmanager, reducing MTTD from 15 minutes to 2 minutes and improving MTTR by 60%.
- Designed and executed large-scale **load testing using k6, JMeter, and custom Kafka stress tools,** validating 350K msg/sec throughput and 15M concurrent connections, identifying 8 architectural bottlenecks prior to production launch.
- Built and containerized a **RAG-based AI assistant using Python (FastAPI), LangChain, Ollama, Llama 3.2, vector databases (FAISS/Chroma), Docker, and REST APIs**, integrating LLM-based contextual search into backend workflows.
- Served as an IC and mentor, leading design reviews and coaching of 5 engineers on distributed systems located globally. Reduced defects by 30% and accelerated delivery through strong code and architecture oversight.

CISCO Systems – Software Engineer **Bengaluru, India**
05/2021 - 08/2024

- Migrated inter-service communication to **gRPC in Golang (grpc-go)**, leveraging HTTP/2 multiplexing and protobuf serialization, reducing request latency by 98% and significantly improving service-to-service throughput.
- Implemented concurrency-safe routing and multi-tenant stream processing in **Go using goroutines, channels, sync primitives, and atomic operations**, achieving 320% faster parallel workload handling under high concurrency.
- Optimized **PostgreSQL/Aurora performance** through composite indexing, query plan analysis (EXPLAIN), connection pooling, and read-replica utilization, reducing query latency by 40%.
- Improved frontend performance by 46% using **React (Vite/CRA), Redux Toolkit, and React Router**, implementing code splitting, lazy loading (React.lazy/Suspense), and memoization strategies to reduce bundle size and improve time-to-interactive.
- Migrated **frontend monolith to a micro-frontend architecture using React, Module Federation, Redux for shared state management**, and independent deployment pipelines, enabling team-level ownership and faster feature releases.
- Implemented sliding-window API rate limiting in **Golang using Redis-backed distributed counters and middleware patterns**, improving system resilience and preventing abuse under burst traffic.

DBS Bank DAH2 – Application Developer Specialist **Hyderabad, India**
05/2020 - 05/2021

- Optimized loan processing with high-performance Java 14 and Spring Boot, leveraging ProtoBuf and Netty for asynchronous operations.
- Created a scalable microservices-based loan system using Spring Cloud, PostgreSQL, and MongoDB, improving the efficiency by 285
- Achieved high throughput and enhanced system communication with Chronicle Queue and RabbitMQ.

Atos-Synel – Associate Consultant **Pune, India**
10/2018 - 05/2020

- Developed and enhanced web application responsible for handling transactions and payments carried out for paying domestic bills.
- Built payment gateway modules and merchant on boarding flows using Spring Boot, Elastic Search(ELK Stack) and Kafka, enabling seamless reconciliation and tax calculation for transaction events.
- Developed reconciliation and settlement microservices with Angular-based UIs.

System Engineer **Pune, India**
Atos *05/2017 - 08/2018*

Java Web Application Developer **Pune, India**
IBM India Pvt Ltd *07/2014 - 05/2017*

Education

B.Tech Information Technology **Bhubaneswar, India**
International Institute of Information Technology *2010 - 2014*

Skills

Languages: Java, Golang, Python
Microservices: Spring Boot, Gin, Fiber, FastAPI
Databases: PostgreSQL, MariaDB, MongoDB, Redis, RocksDB, Cassandra
Frontend: React 18, TypeScript, TailwindCSS, Vite, ShadcnUI
Security: Spring Security, JWT, OAuth2
AI: RAG, LangChain, LangSmith, Llama 3.2, HuggingFace
Cloud & DevOps: Docker, Kubernetes, AWS (EKS, ECS, Lambda), Jenkins, Git

Open Source Projects

(*Github Project titles link to source code*)

- **Payment Service Provider**
  Built a high-scale payment platform using Spring Boot and Kafka, focusing on distributed transactions, event-driven architecture, and scalable microservices design.
- **Real-Time Chat Platform + WebRTC Signalling**
  Designed a production-grade real-time communication system using Go microservices, WebSockets, and concurrency-safe patterns to handle high throughput messaging.
- **Dockerized AI Agent**
  Developed an AI-powered email assistant using LangChain, RAG pipelines, and Llama models, fully containerized with Docker for scalable deployment.